RUSUMBA BUKANGA

Cofounder & Chief Technology Officer | Full Stack Developer | DevOps Engineer

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PROFESSIONAL SUMMARY

Cofounder & Chief Technology Officer and Full Stack Developer with comprehensive experience building and maintaining the digital foundation that powers modern insurance, hospitality, and fitness operations. Currently operating across Alpha Group and Cleverly AI, translating complex business workflows into efficient, secure, and scalable software systems.

My work spans the entire technology ecosystem, from centralized authentication and client-facing booking, membership, and policy management platforms to internal CRM systems, analytics dashboards, and secure API integrations. I architect solutions handling sensitive data while ensuring POPIA and GDPR compliance, system reliability, and long-term scalability.

TECHNICAL SKILLS

Frontend: Next.js, React, HTML5, CSS3, Tailwind, TypeScript, ES6

Backend: Node.js, Express.js, FastAPI, RESTful API Design

AI / ML: LLM integration (Claude, AWS Bedrock), prompt engineering, RAG & embeddings, vendor-neutral AI provider layers, OCR pipelines, inference cost optimization

Auth & Identity: OAuth2/PKCE, AWS Cognito, JWT/KMS-signed tokens, SSO gateway design

AWS: Lambda, EventBridge, API Gateway, Cognito, DynamoDB, S3, KMS, CloudFront, SES, SNS, Secrets Manager, IAM, Bedrock

Databases: MongoDB, PostgreSQL, Supabase, DynamoDB, Google BigQuery

DevOps & Cloud: Docker, Kubernetes, CI/CD Pipelines, GCP, AWS, Azure, Vercel, Render

Security Testing: Kali Linux, Nmap, Metasploit, Burp Suite, vulnerability & dependency auditing, secure coding practices

Testing: Unit & Integration Testing, Jest, Automated QA

Version Control: Git, GitHub, GitFlow

Tools: Webpack, Babel, Gulp, Postman, Jira, Notion

Other: Python scripting, Automation, Network Infrastructure, Security Hardening

PROFESSIONAL EXPERIENCE

Cofounder & Chief Technology Officer | Cleverly AI

South Africa | Oct 2025 – Present

- Cofounded and lead technical direction for an AI-assisted grading platform built for South African schools, colleges, and universities
- Architected a full-stack platform (Next.js, Express, Supabase) with a vendor-neutral AI provider layer, enabling rubric-aligned, evidence-linked grading with teacher-in-the-loop review
- Integrated large language model providers (Claude) into core product functionality, including prompt engineering, retrieval-augmented generation (RAG), and local embedding pipelines
- Built a tiered OCR pipeline combining free text-layer extraction with vision-based handwriting recognition to control cost while preserving grading accuracy
- Implemented prompt caching, model tiering, and cross-provider cost telemetry to keep unit economics sustainable as usage scales
- Led a security hardening pass covering access controls, rate limiting, and POPIA-aligned data handling
- Own end-to-end DevOps: CI pipelines, automated testing, and production deployment across Vercel and Render

- Drive go-to-market strategy, including founder-led content and pilot outreach to South African educational institutions
- Manage engineering hiring, budgets, and vendor relationships; contribute at board level as a contractually appointed Director

Full Stack Developer | Alpha Group

Johannesburg, South Africa | 2025 – Present

Alpha Group businesses include Alpha Insure, The 11th Floor restaurant, and Body Action Gym.

- Designed and built a centralized SSO authentication gateway serving every application across the organization's platform ecosystem | underwriting, client-facing, and internal HR systems | built on AWS Lambda, API Gateway, Cognito, DynamoDB, and KMS-signed JWTs
- Built an internal HR portal with org-chart, employee self-service, events, and announcements functionality, integrated with Microsoft Graph

Alpha Insure (Insurance & Financial Services):

- Develop and maintain internal CRM systems for underwriting, claims, and broker management
- Build customer-facing quote calculators, policy management portals, and secure payment workflows
- Led development of a broker onboarding portal, streamlining agent recruitment and compliance workflows
- Design and deploy RESTful APIs integrating underwriting partners, banks, and reinsurers
- Create real-time dashboards and automated document generation systems
- Reduced policy processing time by 60% and manual data-entry errors by 85%

The 11th Floor (Hospitality & Events):

- Built a digital booking and reservation system for a high-traffic rooftop restaurant
- Developed event and capacity management dashboards

Body Action Gym (Fitness & Wellness):

- Designed and built a membership management and event scheduling platform
- Implemented onboarding, renewal, and engagement workflows
- Built the automated recurring membership debit order run on AWS (Lambda, EventBridge for scheduled batch processing, S3, SES/SNS for notifications), including bank reconciliation matching against member accounts

Chief Technology Officer | 123Tutors (Pty) Ltd

South Africa | Aug 2025 – Present

- Returned to 123Tutors in a Chief Technology Officer capacity, owning technical strategy and architecture for the tutor-student matching platform
- Led a full platform migration from Bubble to Next.js/TypeScript, modernizing the stack onto Prisma, Supabase (Postgres, Auth, Storage), Vercel, and AWS (S3, Lambda, EventBridge for scheduled jobs)
- Led a comprehensive codebase and security review, resolving critical production risks and strengthening platform resilience
- Directed the technical roadmap for the platform's matching engine and new institutional reporting capabilities
- Shipped an AI Tutor feature powered by Claude (via AWS Bedrock) | real-time streaming chat with intent classification, worked examples, and generated interactive mindmaps/diagrams for student learning support
- Own CI/CD, RLS-based authorization design, and third-party integrations (Paystack, Twilio, WhatsApp Cloud API)

NOC Support Engineer | Reflex Solutions (Pty) Ltd

Johannesburg, South Africa | Dec 2024 – Jul 2025 (Contract)

- Provided 24/7 network operations support for enterprise environments
- Monitored, diagnosed, and resolved network and infrastructure incidents
- Maintained and supported FortiGate firewalls, F5 Big-IP load balancers, and Cisco network infrastructure
- Improved fault detection and response times using PRTG and NetBox monitoring tools

- Automated network diagnostics and troubleshooting workflows using Python scripts
- Worked closely with vendors and carriers to restore services and implement scalable solutions
- Applied ITIL-aligned incident, problem, and change management practices

EDUCATION

- BCom, Logistics Management | University of Johannesburg
- SLP, Quantitative Finance | University of Johannesburg

KEY PROJECTS

- Cleverly AI Grader | AI-assisted, evidence-linked grading platform for schools, colleges, and universities (cleverly.co.za)
- Centralized SSO Authentication Gateway | Auth service (OAuth2/PKCE, Cognito, KMS-signed JWTs) serving an organization's full application ecosystem
- Alpha HR Portal | Internal employee self-service platform with org-chart, events, and Microsoft Graph integration
- Broker Onboarding Portal | Streamlined agent recruitment and compliance workflow for Alpha Insure
- Restaurant Booking Platform | Real-time reservation system with dynamic dashboarding
- Body Action Events Portal | Event scheduling and membership management system
- Membership Debit Order Run | Automated recurring billing and bank reconciliation system for gym memberships, built on AWS (Lambda, EventBridge, S3, SES/SNS)
- 123tutors Platform Migration | Led migration from Bubble to Next.js/TypeScript with Prisma, Supabase, and Vercel; architected the tutor-matching engine roadmap (123tutors.co.za)
- AI Tutor (123tutors) | Claude-powered (via AWS Bedrock) streaming chat tutor with intent classification, worked examples, and generated interactive mindmaps/diagrams
- Cloud-Native Recommendation Engine | Deployed FastAPI and BigQuery-based system on GCP
- Network Automation Suite | Created Python automation tools for enterprise diagnostics
- Security Reviews | Conducted internal security and dependency audits on production web platforms, implementing secrets management and secure-coding standards
- Personal Cybersecurity Lab | Built a segmented, VPN-gated home lab for offensive security practice using Kali Linux (Nmap, Metasploit, Burp Suite) against intentionally vulnerable applications (OWASP Juice Shop, DVWA)

CORE COMPETENCIES

Full Stack Web Development, AI/LLM Integration, API Design & Integration, Authentication & Identity Systems, Cloud Infrastructure & Automation, System Architecture, CI/CD & DevOps, Production Operations, Application Security & Vulnerability Auditing, Incident Response, Performance Optimization, Compliance

LANGUAGES

English (Fluent) | French (Fluent)